

This exam has 2 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Axioms for semantic equivalence (7 points)

Show, using the axioms for semantic equivalence, that

$$\neg \perp \equiv \top$$

(Hint: Use the law $\phi \wedge \neg\phi \equiv \perp$.)

2. Logic circuit (7 points)

Construct a logic circuit that corresponds to the formula

$$(p \wedge q) \vee \neg(p \vee \neg r)$$

3. OBDD (2 + 7 + 7 points)

(a) Represent the formula

$$(\neg x \vee y) \wedge (x \vee \neg z)$$

by means of a binary decision tree, with respect to the variable ordering x, y, z .

(b) Reduce the binary decision tree from (a) to a reduced OBDD, using the steps C1,2,3.

(c) Construct an OBDD for

$$\forall y ((\neg x \vee y) \wedge (x \vee \neg z))$$

from the reduced OBDD you constructed in (b).

4. Predicate logic (6 + 9 points)

Consider the following two formulas:

- $\exists y \forall x R(x, y)$
- $\forall x \exists y R(x, y)$

(a) Give a model that shows these two formulas are not semantically equivalent.

(b) Does one of these formulas semantically entail the other? (Explain your answer.)

5. Equivalence relations (6 + 9 points)

In this problem, we consider the relation R in the set $A := \{1, 2, 3, 6, 9\} \times \{1, 2, 3, 6, 9\}$ that is defined by the description

$$\langle a, b \rangle R \langle c, d \rangle \iff \frac{a}{b} = \frac{c}{d}$$

- (a) Show that R is an equivalence relation.
- (b) Explicitly write down all the different equivalence classes of the equivalence relation R in A , and give a complete system of representatives.

6. Functions (5 + 5 + 5 points)

We are given the functions $sqr: \mathbb{R} \rightarrow \mathbb{R}$, $dec: \mathbb{R} \rightarrow \mathbb{R}$ and $inv: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$sqr(x) = x^2 \qquad dec(x) = x - 1 \qquad inv(x) = \frac{1}{x}$$

- (a) For each of the functions sqr , dec and inv , determine whether or not the function is injective. Please give a brief explanation of your answer in each case.
- (b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by

$$f := inv \circ sqr^{-1} \circ dec.$$

- (c) Express the function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by the description

$$g(x) := \left(\frac{1+x}{x} \right)^2$$

as a composition of the functions sqr , dec , inv and their inverses.

7. Induction (5 + 10 points)

We claim that the following statement is true for all integers $n \geq 1$:

$$\sum_{k=1}^n 3k(k-1) = n(n^2 - 1)$$

(recall that $\sum_{k=1}^n a_k$ is shorthand notation for $a_1 + a_2 + \dots + a_n$).

- (a) Verify by explicit calculation that the statement is true for $n = 1$, $n = 2$ and $n = 3$.
- (b) Using mathematical induction, prove that the statement is true for *all* integers $n \geq 1$.